Class 2: Properties of Materials

While we are all familiar with materials due to our day to day usage, it is actually quite surprising to note that we often know very little about these materials.

Take for example a metal. Metals are extremely commonplace. Yet ask yourself the following question:

"Define a metal"

You will find that the answer is not as commonplace as the material itself!

The answer is "Metals are materials with a positive thermal coefficient of resistivity".

What this means is that when the temperature of the metal is raised, its resistance increases. Other materials which can also conduct charge, such as ionic conductors and semiconductors, actually lower their resistance when the temperature increases. Why do these materials behave differently when we raise the temperature? - we will examine this a little later in this course. For now let us examine our knowledge of materials a little more and identify how we typically understand and classify materials.

In reality, it turns out that it is difficult to strictly define any type of material. The general approach is that we look at properties displayed by a material, and if the values of those properties fall in certain ranges, then we call it a certain type of material. This approach to classifying materials is impacted by the fact that properties displayed by materials are often significantly influenced by the environment that they are placed in. Materials that are insulating at room temperature and atmospheric pressure can become metallic under very high pressures. It is therefore important to keep in mind the conditions under which the material is being tested and to classify the materials within the framework of those conditions. In common usage, the conditions under which materials are being tested and classified are typically room temperature and atmospheric pressure. These conditions are not explicitly stated at times, but we should be alert to these regardless.

When we examine material properties, we find that there are interesting correlations between properties. For example, good conductors of electricity are also often good conductors of heat. Such a correlation implies that something is fundamentally common to the process of conduction of electricity as well as heat.

More than simply being aware of material properties, it is of interest to understand properties of materials and recognize relationships between these properties. Such an understanding enables us to see how material properties evolve from the behavior of the constituents of the material such as atoms and electrons.

Figure 2.1 schematically shows the properties we commonly encounter in materials.

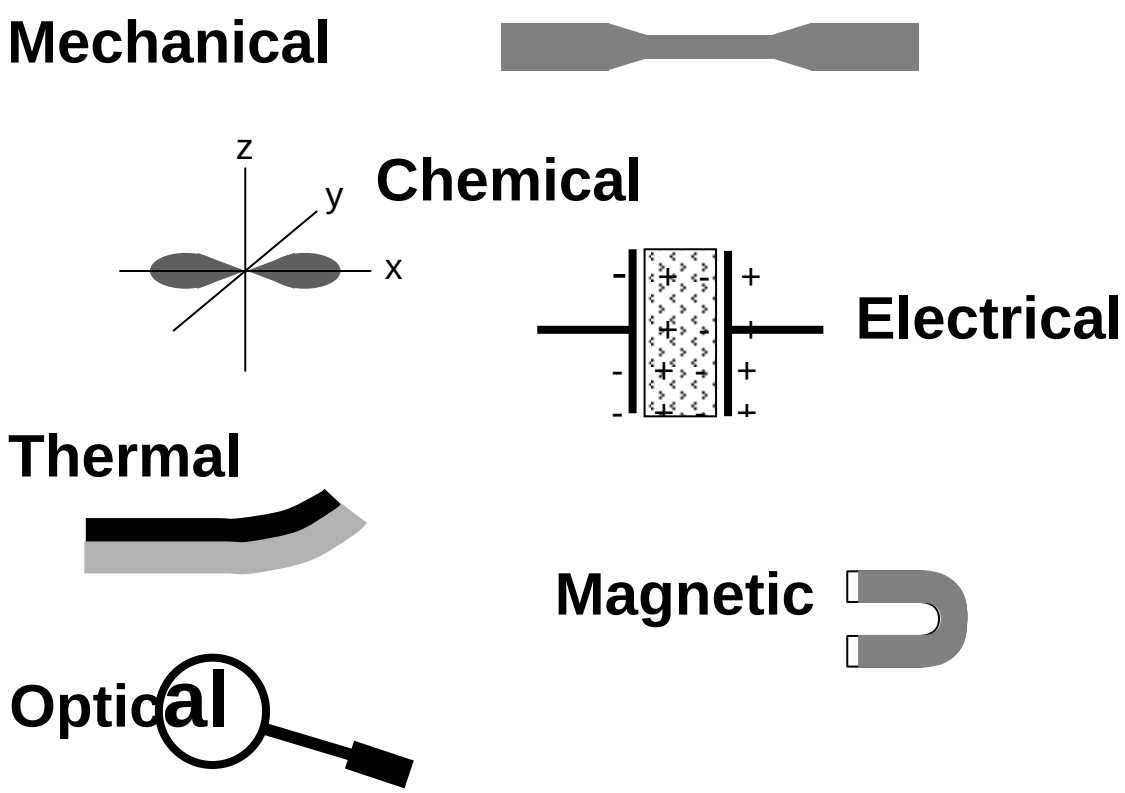


Figure 2.1: Schematic showing the properties commonly encountered in materials.

The material properties that we commonly encounter and some examples of the instances where we encounter them are as follows:

Mechanical properties make their presence felt in a variety of objects we use in terms of the dimensional stability of the object as we use it, the response of the object to the physical forces it is subject to during use or during testing. **Chemical properties** are encountered routinely in the form of cleaning agents we use for example. Cleaning agents typically have instructions on what they can be used for and what they should not be used for. Such instructions reflect the chemical reactivity of the agents. **Thermal properties** are used in commonplace equipment such as in the bimetallic strip in iron boxes, the insulation in thermal wear etc. **Electrical properties** are used extensively in many household entertainment devices. **Magnetic properties** are used in fans, motors, and audio speakers. **Optical properties** are used to create sun control films, and lenses for a variety of applications.

A more elaborate, although not exhaustive, look at these properties is presented below.

Mechanical properties:

The major mechanical properties are shown in figure 2.2:

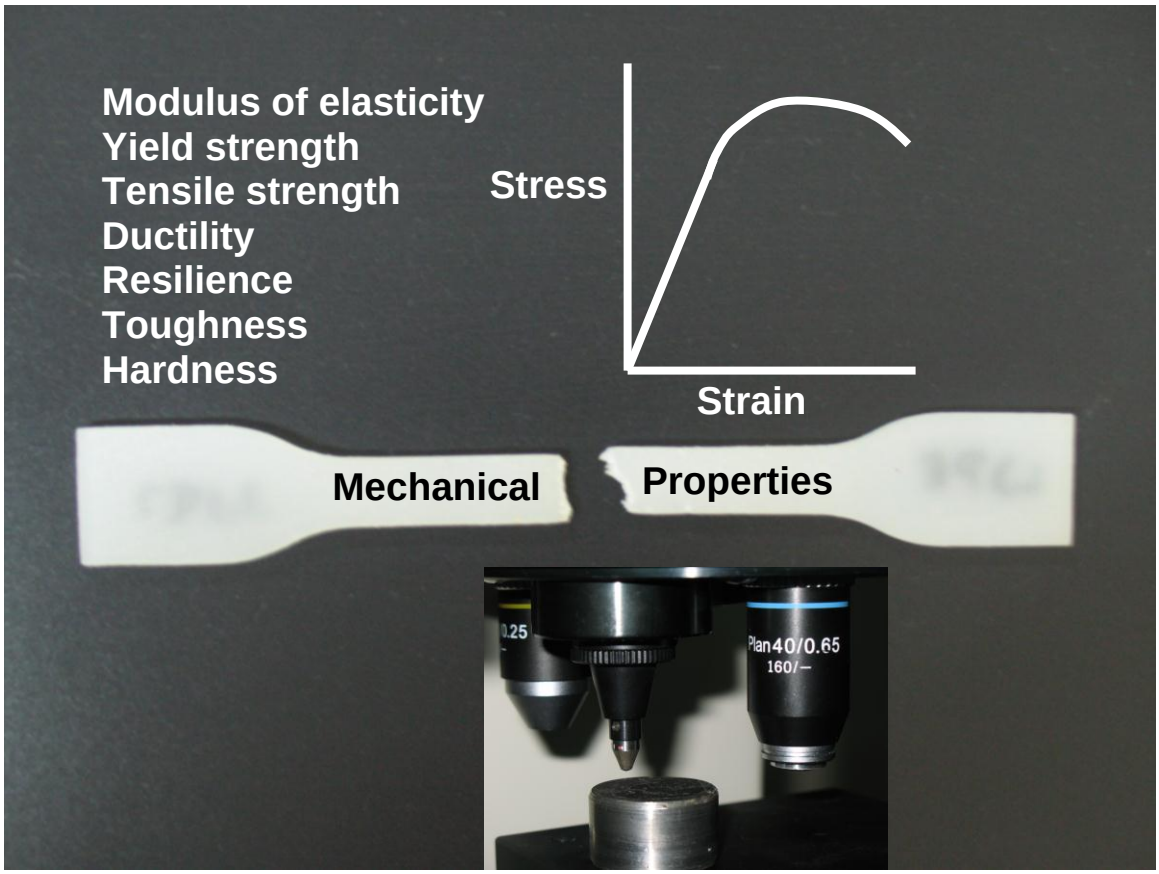


Figure 2.2: A list of the major mechanical properties of a material. Also seen are a 'Stress Vs Strain' curve, from which many of the mechanical properties can be obtained. A dog bone shaped sample that has been tensile tested to fracture is shown. Also visible is a hardness tester in the bottom of the figure.

A tensile test, during which a dog bone shaped sample is pulled till it fractures, enables us to obtain a Stress Vs Strain curve for the material.

From the stress-strain curve, the Modulus of elasticity is obtained as the slope of the linear region of the curve.

The yield strength is the stress at which the stress-strain curve just begins to lose its linearity.

Tensile strength is the maximum stress supported by the sample and is the highest stress recorded in the stress-strain curve.

Ductility is the elongation that can be sustained by the sample before it fractures.

Resilience is the area under the stress strain curve corresponding to only the linear region of the curve. It represents the amount of energy that the material can absorb and still not deform plastically.

Toughness is the total energy absorbed by the material till it fractures and is obtained as the total area under the stress strain curve till fracture.

Hardness is the ability of the material to resist deformation when subject to a local compressive stress.

Mechanical properties are largely a result of material details at the microscopic level. The bonding present between the atoms in the material and the crystal structure adopted by the material, and hence its slip systems, directly impact the mechanical properties. While mechanical properties are measured in a macroscopic level, they originate from the atomic and crystal structure level.

Chemical properties:

We are generally aware that chemical properties are related to details at the atomic level. Figure 2.3 indicates a few of the important chemical properties.

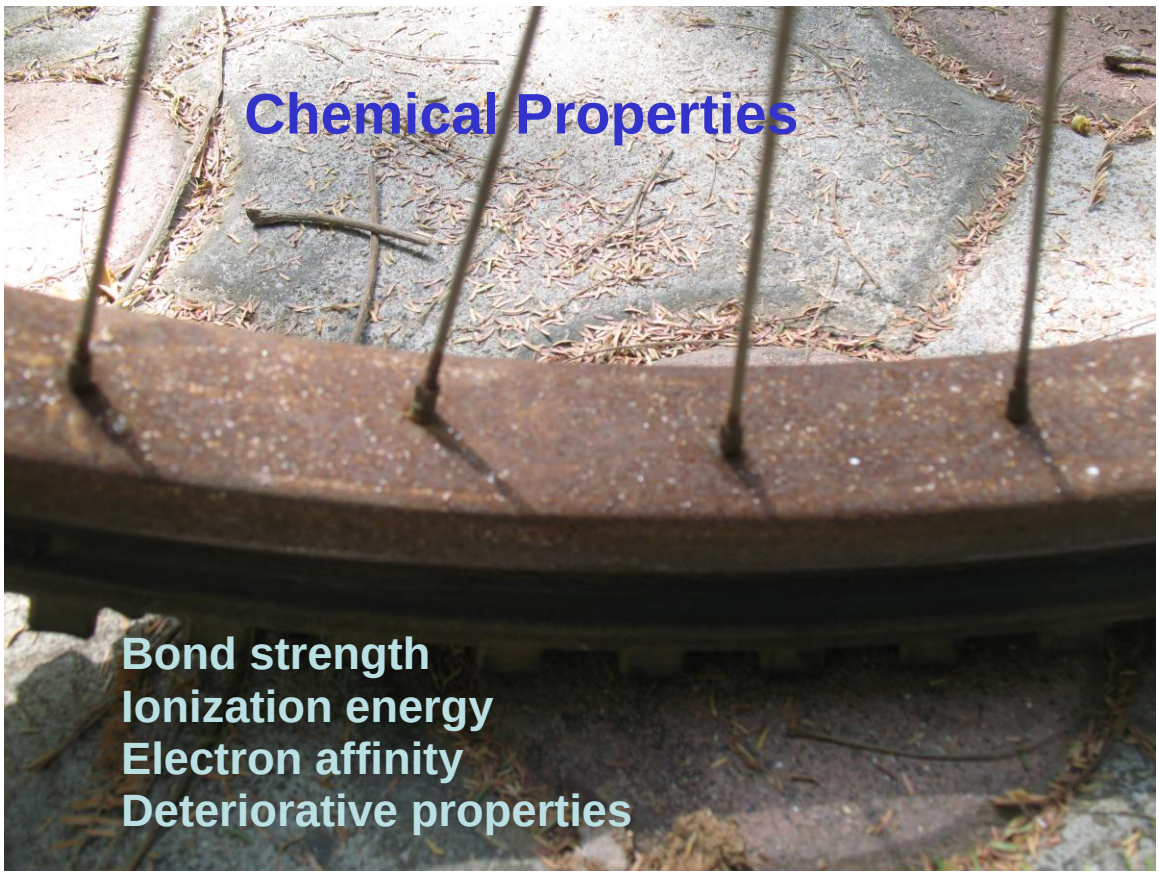


Figure 2.3: A list of the important chemical properties. In the background is a picture of a rusting bicycle wheel, a result of chemical properties of the material used for the wheel.

Bonding energy, ionization energy, and electron affinity, represent quantities associated with exchange of subatomic particles, specifically electrons, between atoms. Taken together they represent the chemical reactivity of the material and indicate, for example, the deteriorative characteristics of the material such as rusting.

Electrical properties:

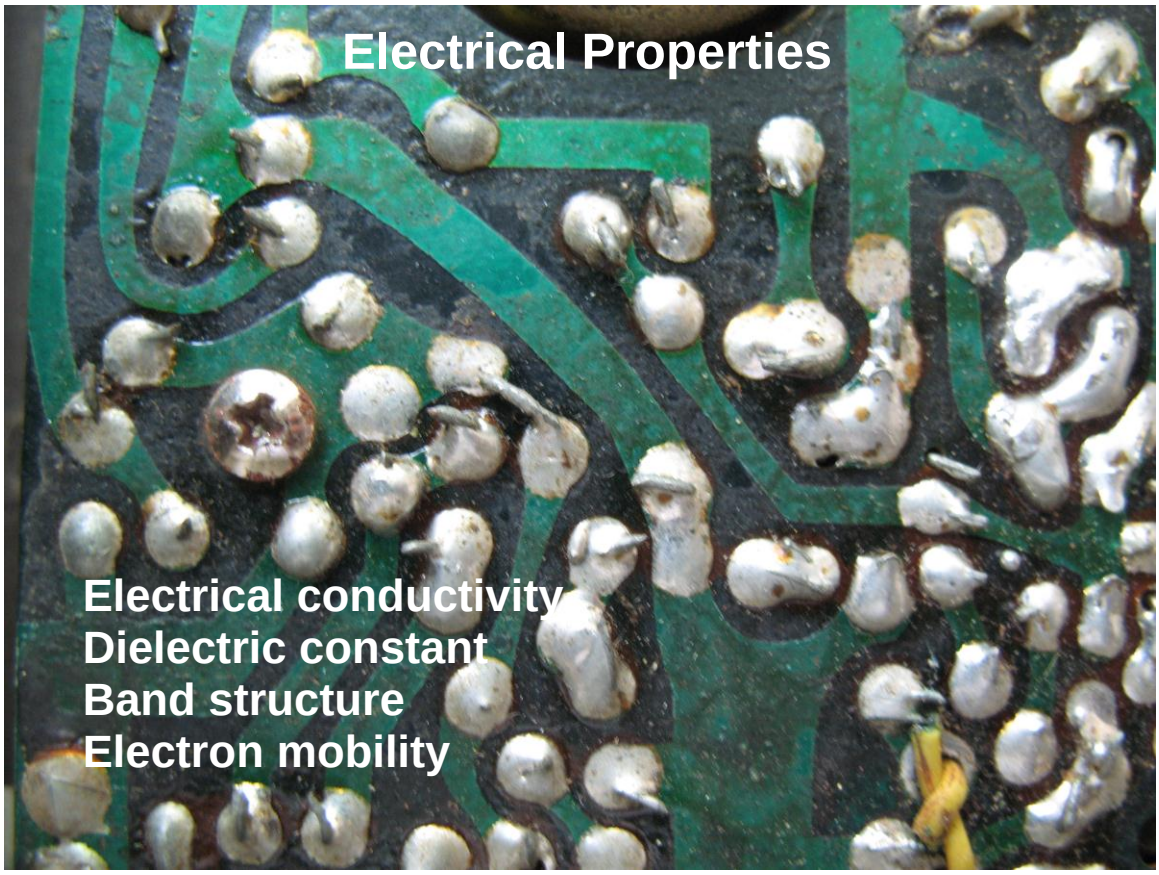


Figure 2.4: A list of the major electrical properties. In the background is part of an electronic circuit.

Figure 2.4 indicates the major electrical properties of materials. Electrical conductivity, or more generally conductivity, represents the transport of charge. Different charge carrier species can be present, such as electron, holes, or ions, and hence we can obtain conductivity with respect to each of these species. It is important to note that simply connecting two different materials which have different charge carriers, will not complete a circuit even if the conductivities of the two materials are the same, because the conductivities are defined with respect to the specific charge carrier in that material. At

the interface between the two materials, neither charge carrier will be able to crossover, if it is not the conducting species in the other material.

Wires connecting devices have electrons as charge carriers, while oxygen sensors in automobile exhausts usually contain oxygen ion conducting materials. Some materials may conduct more than one species.

It is also important to understand the origin of other electrical properties such as dielectric constant, and band structure. It is of interest to see how constraints placed on the constituents of the materials, such as atoms and electrons, result in these properties for the material.

Electron mobility represents the ease with an electron moves in the presence of a field, and indicates the velocity that the electron will attain in the presence of a unit field. Mobility is also used to describe the movement of other species such as atoms, in the presence of an appropriate driving force, and is hence used for diffusion related phenomena as well. In other words, phenomena at electronic as well as atomic level show similarities to each other.

The circuit in Figure 2.4 also shows signs of corrosion, which is an electrochemical phenomenon. Corrosion can degrade a circuit and limit its useful lifetime, and represents an interaction of electronic and chemical properties.

Thermal properties:

Good conductors of heat are also usually good conductors of electricity. Figure 2.5 lists the major thermal properties of materials. The figure also shows a machine that is covered with snow. Such machinery may be required to start in cold weather. The temperature outside could be well below 0 ^{o}C, while the temperature inside the engine could be around 1000 ^{o}C. The materials involved should be able to handle such temperature gradients.

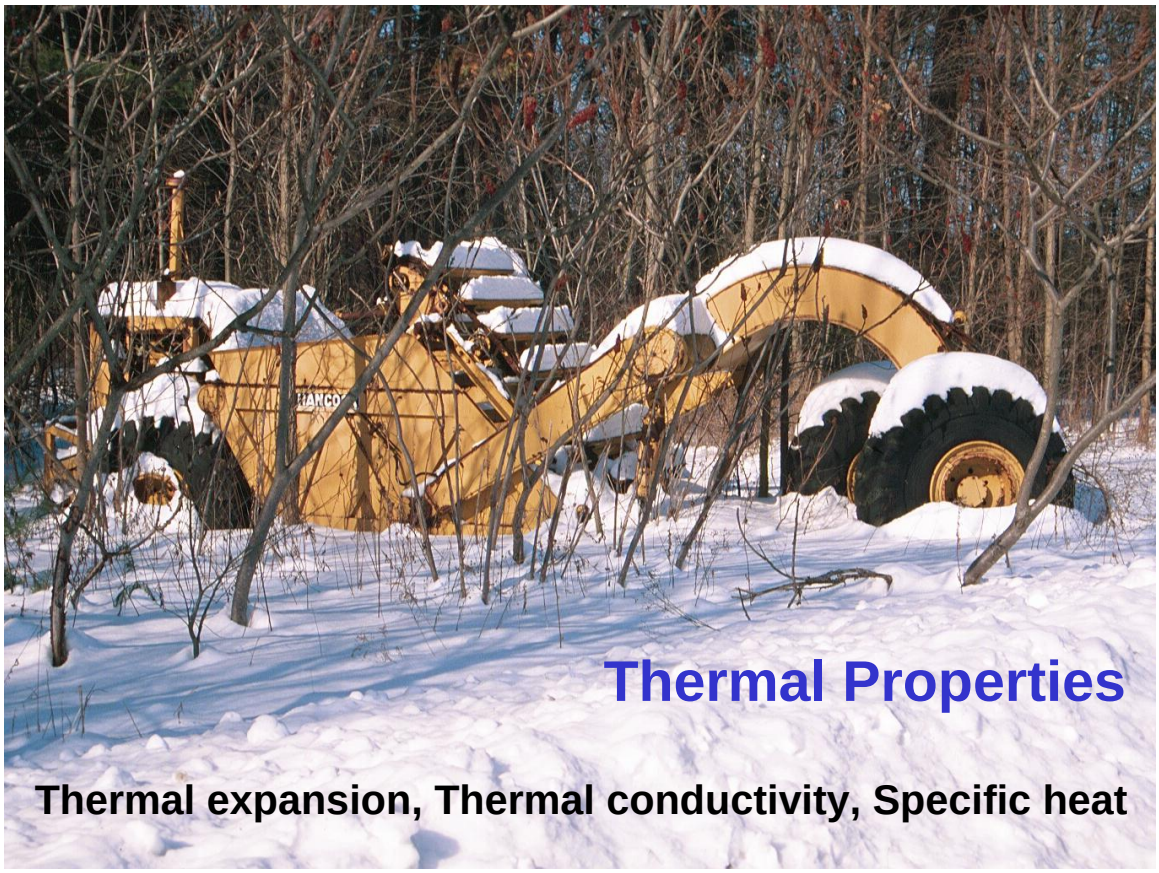


Figure 2.5: A list of the major thermal properties of materials. In the background is a machine that is covered with snow and may be expected to start and operate in cold weather.

Why do some materials expand significantly while heating and some others don't? Thermal expansion is considered in great detail in the next class, and addresses this question. Thermal expansion is associated with phenomena at the atomic level.

Thermal conductivity as well as specific heat are phenomena that have contributions at the atomic as well as electronic level. Based on the material and the circumstance, either the atomic or the electronic contribution to these properties can dominate.

Magnetic properties:

A variety of parameters describe the magnetic properties displayed by materials. Figure 2.6 lists the major magnetic properties displayed by materials.

The Curie Temperature is the temperature above which a ferromagnetic material displays paramagnetic behavior.

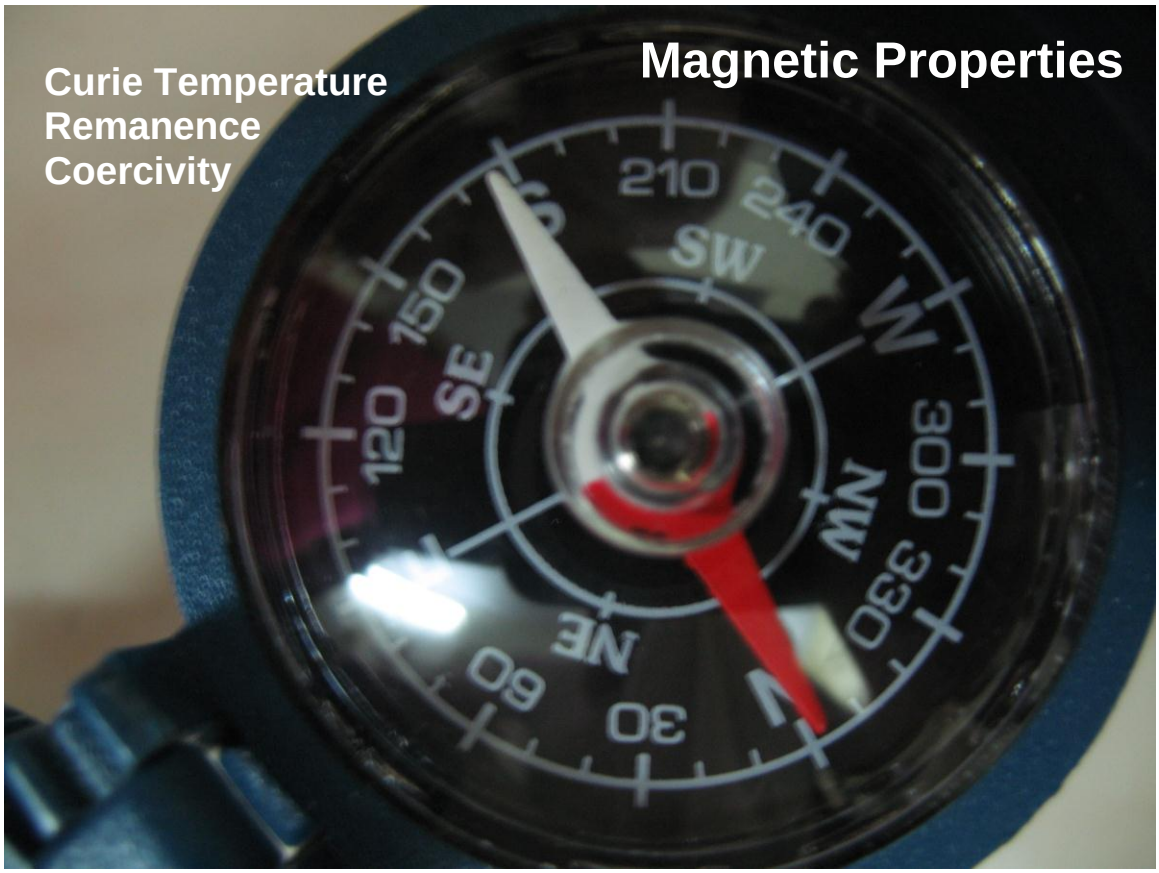


Figure 2.6: A list of the major magnetic properties displayed by materials. In the background is a compass, which uses magnetic properties to carry out its function.

Materials respond in different ways to the presence of an externally applied magnetic field. This response leads to the display of properties such as remanence and coercivity. It also leads to the display of hysteresis. Based on the extent of the hysteresis, the materials are classified as Soft magnetic materials, which show very little hysteresis, or Hard magnetic materials, which show large amounts of hysteresis.

The theories describing magnetism are quite involved and are explored to some degree later in this course.

Although the theories are quite involved, the use of magnetism is relatively common place. In addition to the compass seen in Figure 2.6 above, audio speakers, fans, and electric motors contain magnets, which are essential to their functioning. In fact, magnets in some of these equipment can affect the performance of other equipment near them, and hence some equipment come with magnetic shielding.

In the medical field, strong magnets are used in specific diagnosis equipment such as MRI (Magnetic Resonance Imaging) equipment. For such strong magnets, superconducting wires are required, an aspect that is explored in greater detail later in this course.

Optical properties:

Sun control films that are used in automobiles and in the construction industry, rely on the reflectivity, absorptivity and transmitivity of materials, which are optical properties of those materials.

Figure 2.7 lists the major optical properties of materials.

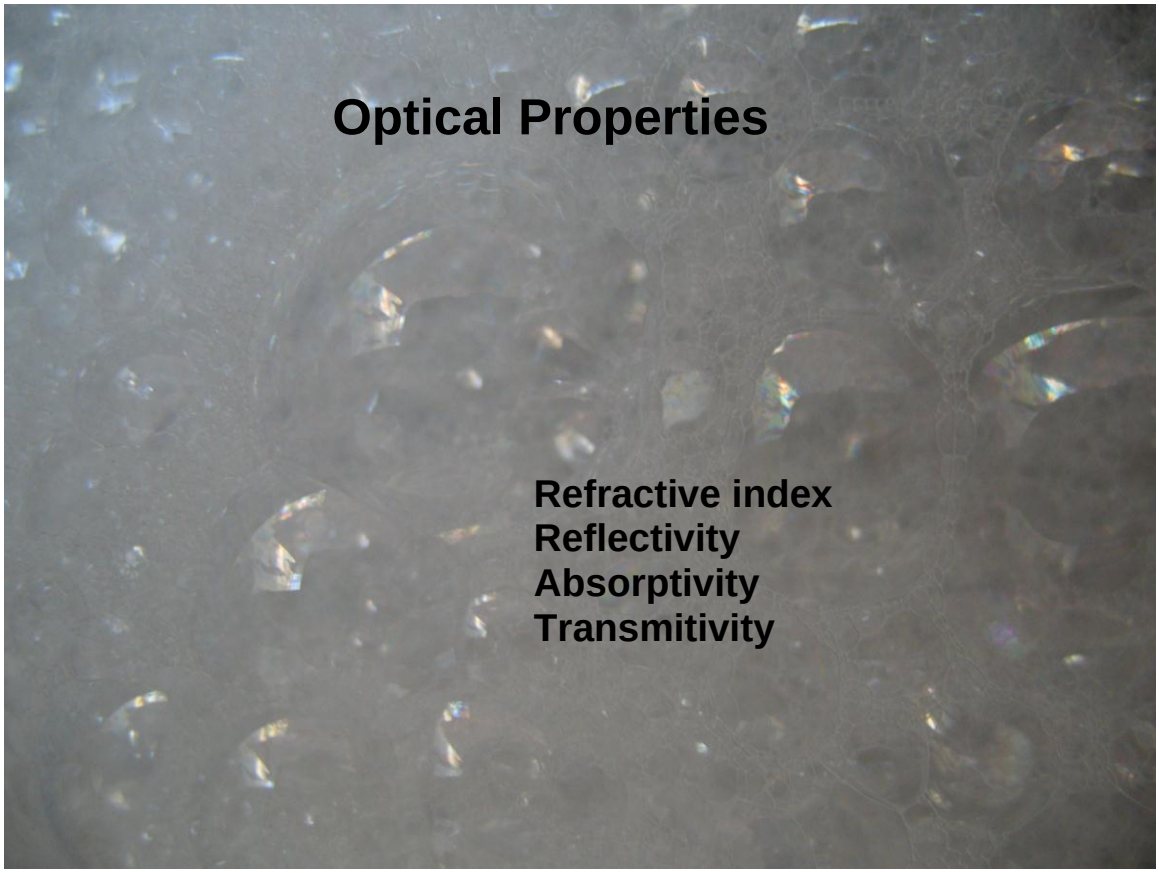


Figure 2.7: List of the major optical properties of materials. The background shows soap bubbles, which display a range of optical properties.

The refractive index is a very important and fundamental optical property. Use of materials for optical purposes is often based on its refractive index.

While carbon in the form of graphite is opaque, carbon in the form of diamond is transparent. Optical properties, just like electronic properties, depend significantly on the electronic structure of the materials. Photoelectric effect is a result of electronic as well as optical properties of the material, and helps us relate the two properties to fundamental phenomena in the material.

It is interesting to note that while lenses can be obtained very cheaply, and can even be made by filling curved surfaces with a transparent liquid such as water, high end photographic equipment can be very expensive. The high cost of such equipment is partly due to the quality of glass used, and also in part to the special surface films that are applied to those glasses to ensure minimal to no additional reflections from the interfaces between the lenses and the atmosphere.

Understanding material properties:

Having discussed material properties in some detail in this class, a brief note on the approaches used to understand the origin of these properties, is presented below.

In Physics books, discussion of properties involves reference to the band structure of the solids and understanding the origin of bands.

In Chemistry books the same topics are often described using atomic and molecular orbitals. Attention is drawn to terms such as Highest Occupied Molecular Orbital (HOMO) and the Lowest Unoccupied Molecular Orbital (LUMO).

While the terminology used may seem different, they are simply different approaches to describe the same phenomena. Choice of approach is often based on convenience and familiarity.

Summary:

In summary, materials display many properties. These properties are often interrelated. It is of interest to understand the origin of these properties. To understand the origin of properties, it will be necessary to start with the constituents of materials such as atoms and electrons, and assign properties to them, and therefore create a model for the material. Based on the properties assigned to the constituents of the materials, it will be possible to predict the macroscopic properties displayed by the materials and to predict the inter relationships between the properties. The effectiveness, with which the predictions match experimental data, will be the measure of the success of the model.